Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering three-dimensional fluorescence from fixed-focus images requires measurements that distinguish structures at different depths. We learn a phase plate for angle-scanned excitation by optimizing depth discrimination independently of reconstruction. A plane wave illuminates a fixed, defocused plate from varying angles while the specimen and detection focus remain stationary. Training minimizes weighted squared correlations between detector-integrated depth signatures, without specimen examples or a reconstruction solver. We evaluate the same plate with total variation (TV) and plug-and-play (PnP) reconstruction using pretrained and neuron-trained denoisers. In simulated mouse pyramidal neuron volumes at eightfold measurement compression and matched nominal dose, axial-profile Wasserstein error falls by 28–50 % and centroid error by 30–54 % relative to the random initialization. Supporting controls show lower centroid error than additional random plates and amplitude masks under pretrained PnP. The objective gives 13–20 % lower profile error than the tested coherence and Fisher-information surrogates under matched training budgets. In controlled crossings, the qualifying gap falls from 3.84 *to* $2.56\,\mu\mathrm{m}$ *with TV and pretrained PnP, but is unchanged with neuron-trained PnP. Learning the excitation code therefore improves axial recovery across the tested priors without including them in optical training.*

1. Introduction

Three-dimensional fluorescence microscopy must distinguish structures that overlap in projection but occupy different depths. When their measurement responses are similar, a sharp reconstruction may still place fluorescence at the wrong axial locations. Reconstruction priors help select a volume, but their success depends on the depth information present in the images [2, 7]. We investigate whether learning the excitation illumination can improve axial recovery across reconstruction methods.

Our system encodes depth through angle-scanned excitation. A plane wave illuminates a fixed, defocused phase plate, producing a spatially structured field in the specimen. Changing the incident angle shifts the excitation differently at different depths, so structures at different axial positions experience distinct sequences of illumination. A camera records one fluorescence image per angle at a fixed detection focus. Only the input angle changes; the specimen, plate and detection focus remain stationary.

We learn the phase plate by reducing similarity between the measurement responses of voxels at different depths and the same lateral position. Each response includes the entire imaging chain: structured excitation, fluorescence detection and camera-pixel integration. The objective uses these physical responses directly, requiring neither specimen examples nor a reconstruction network. It builds on coherence-based sensing design and decoder-independent optical optimization [6, 18, 25], specializing them to excitation-side depth encoding. Unlike training through a chosen decoder [11, 15], this formulation leaves the reconstruction prior outside optical training.

We evaluate the same learned plate on simulated mouse pyramidal neuron volumes using TV and PnP with pretrained and neuron-trained denoisers. Axial-profile and centroid errors quantify depth recovery, complemented by image quality. Additional random-plate and amplitude-mask controls test the baseline choice, while controlled crossings assess separation of overlapping structures. We also compare optical criteria and examine measurement, noise and alignment budgets.

Our contributions are:

1. **Excitation-side axial encoding.** We develop a fixed-focus imaging model in which an angle scan through a defocused phase plate encodes depth, with vectorial propagation and voxel-integrated detection responses.
2. **Learning from optical responses.** We optimize the plate using a weighted squared-correlation objective on depth signatures, without specimen examples or a reconstruction solver.
3. **Evaluation of axial recovery across priors.** We test the same learned optics with TV and two PnP decoders, evaluating full depth profiles, centroid accuracy and controlled structural separation under matched acquisition conditions.

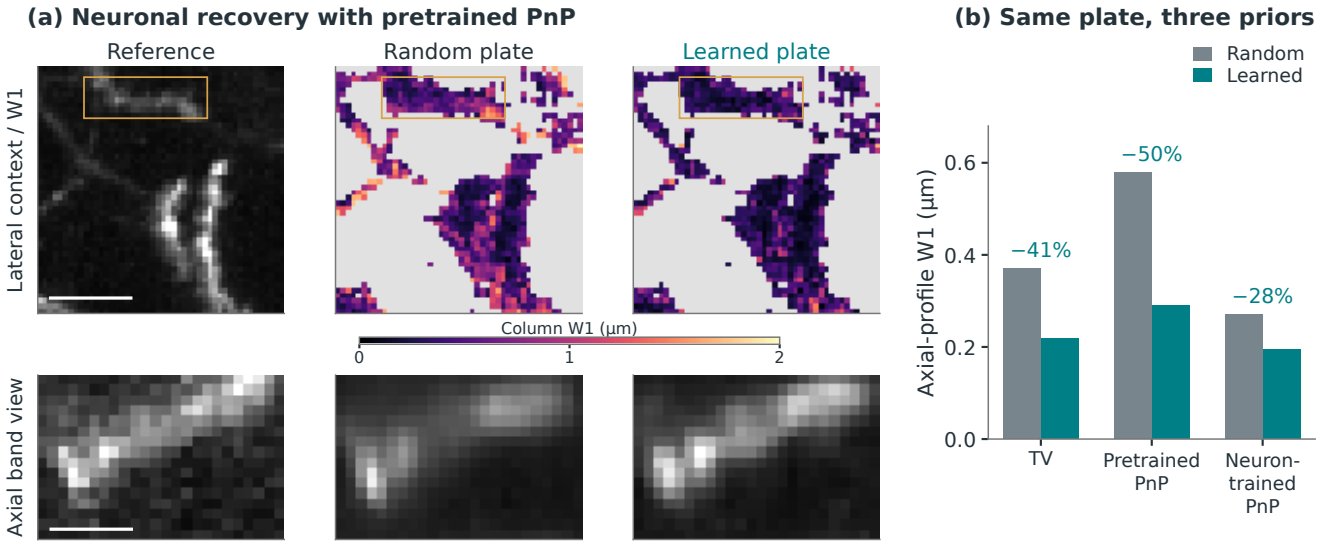


Figure 1. **Learning the illumination improves axial recovery across priors.** (a) Reference projection and column-W1 maps above; matched axial band views below ($C = 8$, draw 0). Pretrained PnP, the prior with the largest mean gain, is shown. The crop and band were selected from reference structure. Band views maximize over the outlined $5.1\,\mu$m strip with true aspect and depth downward; one linear intensity scale is shared below. Gray marks excluded columns in the error maps. (b) Mean W1 over 60 crops using the same learned plate. Centroid errors and paired intervals: Tab. 1. Bars: $10\,\mu$m (reference projection), $5\,\mu$m (band view).

2. Related work

Depth encoding in fluorescence microscopy. Computational fluorescence microscopes encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D recover volumes from depth-dependent detection responses [1, 4, 25]. On the excitation side, Yeh *et al*. [26] combine translated speckles with axial scanning. Pascucci *et al*. [16] exploit axial speckle decorrelation for volumetric recovery from lateral scans, with fluorescence saturation improving resolution. DEEP and DEEP^2 combine excitation patterning with computational fluorescence recovery through scattering [24, 29]. Angular scanning through a scattering layer has also enabled two-dimensional fluorescence recovery [27], while tilted pattern projection produces depth-dependent lateral shifts [9]. We learn a defocused phase plate for angular excitation scanning without axial specimen motion or detection refocusing.

Learning optical encoders. End-to-end methods optimize optics jointly with reconstruction networks [8, 15, 22]; Kellman *et al*. [11] optimize illumination through a specified reconstruction algorithm. Decoder-independent design instead uses properties of the measurements. Shechtman *et al*. [21] optimize Fisher information for emitter localization, and Miniscope3D [25] reduces cross-coherence between detection PSFs across depth. Gram-matrix and coherence objectives also have a long history in sensing-matrix design [5, 6]. IDEAL [18] optimizes estimated mutual information using fitted measurement-distribution models. IDEAL-IO [14] alternates density fitting with optical optimization, including for snapshot 3D fluorescence microscopy. Our method belongs to this measurement-driven family. Its objective compares detector-integrated voxel responses across illumination angles, targeting different depths at each lateral location. It requires no specimen examples, fitted specimen distribution or reconstruction solver.

Volumetric reconstruction and priors. TV promotes piecewise-smooth structure [20], while PnP methods use denoisers as implicit priors [10, 23]. Multi-slice fusion combines denoisers along complementary orientations of a shared volume [12]. We adapt this framework to fluorescence reconstruction, testing both pretrained and neuron-trained priors alongside TV. This evaluates optical gains under different prior assumptions.

3. Optical model and learning

Acquisition. A tilted plane wave illuminates a fixed phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, where $\mathbf{r} = (x, y)$. For direction t, its incident field is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^{\top}\mathbf{r}}$, with transverse direction cosine $\mathbf{s}_t$ and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ in the simulated medium of refractive index one. The supplement specifies the 256-direction scan. After phase modulation, angular-spectrum propagation carries the field over distance D to an intermediate illumination image plane. The objective pupil then filters this field before a vectorial Richards–Wolf

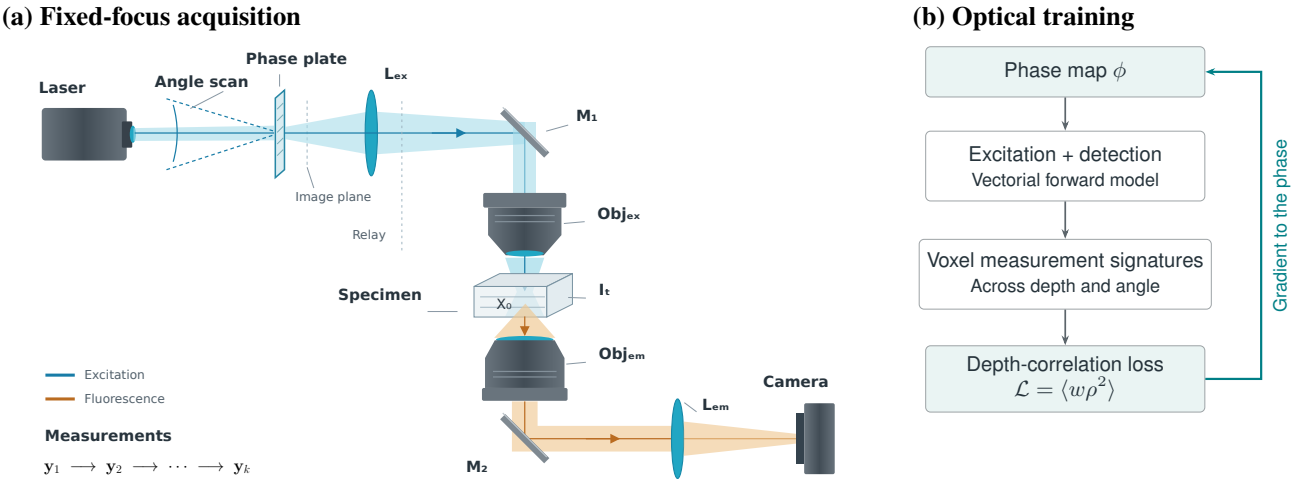


Figure 2. **Fixed-focus acquisition and optical training.** (a) An angle-scanned plane wave passes through a defocused phase plate; the camera records fluorescence at a fixed focus. Blue and orange indicate excitation and emission. (b) The plate is optimized through the optical model using depth-signature correlations. Reconstruction is evaluated separately.

relay [13, 19] produces the specimen excitation, $I_t(\mathbf{r}, z) = \sum_{c\in\{x,y,z\}} |E_{t,c}(\mathbf{r}, z)|^2$. The vectorial model accounts for the high excitation NA; the incident polarization is along x. All coordinates are specimen-equivalent. The fixed-focus camera records one image per direction, so acquisition is sequential.

Why the angle encodes depth. Propagation after the phase plate converts its phase structure into spatially varying intensity. A change in illumination angle shifts this pattern differently at different depths. In the ideal paraxial thin-screen model, changing the transverse direction cosine by $\Delta\mathbf{s}$ gives a lateral displacement $\Delta\mathbf{r} \approx d\,\Delta\mathbf{s}$ after propagation over distance d. Each depth therefore experiences a different sequence of excitation intensities during the scan. Defocusing the plate places the shear pivot outside the specimen. This relation explains the encoding principle; the vectorial relay and finite pupil determine the actual patterns, including departures from pure translation.

Image formation. Let $X_0(\mathbf{r}, z)$ be the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$ [26]. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z')X_0(\mathbf{r}', z')I_t(\mathbf{r}', z')\,\mathrm{d}^2\mathbf{r}'\,\mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity PSF and Ω is the specimen volume. Pixel p, occupying region D_p, records $\bar{y}_{t,p} = |D_p|^{-1}\int_{D_p} g_t(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$. Fluorescence adds incoherently before pixel averaging. Excitation multiplies the specimen before detection blur, allowing fine illumination structure to affect measurements even when the camera cannot resolve that pattern directly.

Voxel-integrated operator. The optical field can vary within a reconstruction voxel. We therefore simulate the optics on a finer grid and integrate its responses over each reconstruction voxel. If B is the fine-grid measurement operator and U replicates each reconstruction coefficient over its fine-grid cells, the reconstruction operator is $\mathsf{A} = \mathsf{BU}$. This is an exact restriction of the discretized model to a piecewise-constant basis.

For a fine-grid specimen $\mathbf{x}_f$ and its voxel-averaged representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$,

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{UR}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Measurements are generated before voxel averaging, retaining the sub-voxel specimen residual $\boldsymbol{\delta}_{\mathrm{disc}}$.

Linear fluorescence permits efficient operator assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells and $g_{p,i}$ is cell i's detection response, including its integration weight. The responses $g_{p,i}$ are computed once and reused across angles. This factorization makes the fine-grid optical model practical to assemble; the implementation cost is detailed in the supplement.

Learning the plate. At lateral position $\mathbf{r}$ and depth z, the corresponding column of A describes how a unit voxel con-

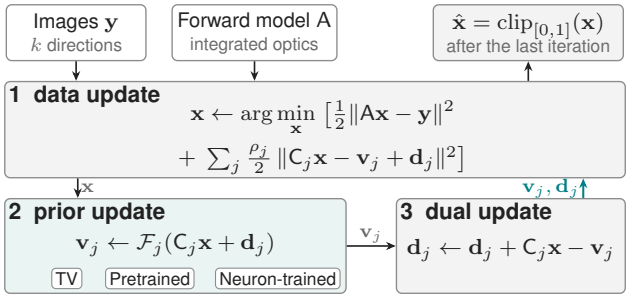


Figure 3. **Reconstruction with interchangeable priors.** A quadratic data update couples the images and forward model to a shared volume. Prior and dual updates feed back through ADMM. Each run uses one prior family; the networks remain fixed across encoders. Branch transforms C_j are identities for PnP and the gradient/identity for TV. TV includes a box-projection branch; PnP clips the final estimate.

tributes to all camera measurements. Let $\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z}$ be this signature restricted to a sampled angle set $\mathcal{T}$, centered over its angle–pixel entries and normalized to unit norm. We minimize

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}}\left[\frac{\sum_{z\neq z'} w_{zz'}\left((\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z})^\top \mathbf{a}^{\mathcal{T}}_{\mathbf{r},z'}\right)^2}{\sum_{z\neq z'} w_{zz'}}\right]. \quad (4)$$

For localized, overlapping detection footprints, centering over angle–pixel entries can leave a common angular component. The loss encourages depth-selective responses that reduce its relative contribution. The fixed weights give selected reflected-depth pairs twice the weight of other pairs; their indices are specified in the supplement. Squaring penalizes correlation of either sign. Signature normalization makes the loss invariant to uniform optical-power scaling, while deposited excitation dose is matched separately at evaluation.

Training differentiates through the excitation, relay and detector model, without specimen examples or a reconstruction solver. We evaluate this depth-discrimination surrogate with the decoders of Sec. 4.

4. Reconstruction with fixed priors

We evaluate the same learned plate with three reconstruction methods: TV, pretrained PnP and neuron-trained PnP. None participates in optical training. Each method uses the operator for its encoder, with the prior held fixed across encoders. Scalar reconstruction parameters are selected separately for each encoder and measurement budget on held-out validation volumes.

Total variation. The non-learned baseline minimizes $\frac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda\|\nabla\mathbf{x}\|_{2,1}$ subject to $\mathbf{x} \in [0,1]^N$, where $\|\nabla\mathbf{x}\|_{2,1}$ is isotropic three-dimensional TV [20]. We solve this problem using ADMM [3].

Plug-and-play reconstruction. PnP combines the measurement model with denoisers acting as implicit priors [10, 23]. Following multi-slice fusion [12], we denoise complementary slices of a shared volume. For orientation $o \in \{xy, yz, zx\}$, P_o rearranges the volume into slices processed independently by a two-dimensional denoiser, giving $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$.

Pretrained PnP uses a natural-image DnCNN [28] along the three orientations, giving $\mathcal{J} = \{xy, yz, zx\}$ without training a 3D network. *Neuron-trained PnP* replaces that 2D denoiser and adds a neuron-trained volumetric branch, augmenting $\mathcal{J}$ with 3D. It therefore changes both the training distribution and volumetric prior.

PnP-ADMM alternates a quadratic measurement-consistency update with these denoisers. Let ρ_j be branch penalties, $\mathbf{v}_j$ auxiliary volumes and $\mathbf{d}_j$ scaled dual variables. The volume update is

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}}\Big[\frac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \sum_{j\in\mathcal{J}} \frac{\rho_j}{2}\|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2\Big], \quad (5)$$

solved by conjugate gradients. Each branch then applies its denoising operator $\mathcal{F}_j$:

$$\mathbf{v}_j^{k+1} = \mathcal{F}_j(\mathbf{x}^{k+1} + \mathbf{d}_j^k),$$
$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}. \quad (6)$$

These are PnP updates; a generic denoiser need not be the proximal map of an explicit regularizer.

An optional ℓ_1 branch is weighted by validation and is inactive in every main-comparison setting. Neuron-trained networks use a separate animal; all network weights remain fixed during reconstruction. Both PnP variants start from normalized back-projection and clip the final estimate to $[0,1]$. TV includes an internal box-projection branch. Architectures, penalties, iteration budgets and parameter selection are specified in the supplement.

5. Experimental setup

Simulation settings. We simulate linear fluorescence in a homogeneous, non-scattering medium. Excitation and emission use a common wavelength of 532 nm as a modeling approximation, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has 0.32 μm features and lies 6.6 μm before the intermediate illumination image plane in specimen-equivalent coordinates. Fields are evaluated at 0.08 μm lateral and 0.32 μm axial spacing, then binned to a 0.32 μm analysis grid. The reconstruction uses 0.64 μm

isotropic voxels. Measurements are generated on the analysis grid before reconstruction-voxel averaging, as in Eq. (2).

Volumes and measurements. Each reconstruction contains $16 \times 48 \times 48$ voxels ($N = 36{,}864$), spanning $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image has 12×12 samples. Acquiring $k = 256$, 64 or 32 illumination directions gives measurement compression $C = N/(144k) = 1$, 4 or 8. For $C > 1$, we use five randomly drawn direction subsets, paired across encoders.

Data and validation. Reference crops come from public two-photon stacks of mouse pyramidal neurons [24], min–max normalized and resampled to the analysis grid. Six cells from three animals contribute 60 test crops: two per cell selected by reference structure before reconstruction, plus eight sampled at random after parameters were fixed. Other experiments use the 12 prespecified crops unless stated otherwise. Denoisers use 459 training crops from a fourth animal. Scalar parameters are selected on two vascular crops from another animal, with no test volume used for training or tuning.

Optical controls and dose. Controls use three random phase plates, three defocused binary masks, a plate-free scan with a $4\times$ aperture and an in-focus DMD with random binary patterns. The relay, pupil, detector and operator assembly are shared. Excitation is simulated beyond the specimen field to suppress beam-edge encoding. Each encoder is normalized to the same mean excitation over the specimen and all 256 directions. With k directions, the photon model is $n_i \sim \mathrm{Poisson}(\kappa_0\, 256\bar{y}_i/k)$. Exposure κ_0 is set separately for each specimen using the random initialization to give 1000 expected photons per camera pixel per image at the mean signal for $k = 256$, then shared across encoders. This matches deposited dose for the full scan and in expectation over random subsets; we call it *matched nominal dose*. Two optical sampling seeds start from the same random phase. The supplement specifies the aperture, noise variants and tuning grids.

Axial and image-quality metrics. Our primary depth metric is the axial Wasserstein-1 error, which compares complete intensity profiles [17]. At each reference-signal column c, we normalize the reference and nonnegative reconstructed profiles to unit mass, with cumulative distributions F_c and $\hat{F}_c$. Then

$$E_{\mathrm{W1}} = \sum_c \pi_c\, \Delta z \sum_{i=0}^{Z-2} |F_c(i) - \hat{F}_c(i)|, \tag{7}$$

where $\Delta z = 0.64\,\mu\mathrm{m}$ and π_c is proportional to reference column brightness, summing to one. We include columns brighter than 10 % of the reference maximum. Centroid error is the corresponding weighted absolute difference between axial intensity centroids. W1 detects profile differences that a centroid can conceal, but does not measure intensity or lateral fidelity; we therefore also report PSNR and foreground SSIM.

Paired comparisons. Encoders are paired on specimens, direction subsets, nominal dose and noise protocol. Draws are averaged within each crop; changes compare the resulting means. We resample the six cells with their crops intact (10,000 bootstrap replicates). These intervals account for within-cell dependence but remain approximate for six cells from three animals.

Controlled structure tests. We also simulate crossing fluorescent filaments over a range of axial gaps, using three lateral crossing geometries, two axial centers and five measurement draws. Successful recovery requires matched axial peaks in columns carrying at least half of the reference crossing signal, together with correct depth ordering of the arms. We report the smallest tested gap at which success reaches 80 % at both centers and persists for larger gaps. A complementary measurement-only test compares neuronal volumes with alternatives preserving their lateral projections and column centroids while merging selected axial profiles. Full definitions are in the supplement.

6. Results

6.1. Depth discrimination before reconstruction

Before learning, angular scanning through the defocused phase plate produces depth-dependent shifts with a pivot near its conjugate plane. Estimated shifts follow the geometrical trend in the finite-pupil model. With 32 directions and unknown emitter brightness, random plates misassign depth in 1.5–1.7 % of point-emitter tests, versus about 5 % for amplitude masks and 25 % for the tested in-focus DMD. Amplitude masks remain competitive with random phase plates in neuronal centroid error. The supplement reports all realizations and defocus settings.

Learning concentrates strong responses into fewer, depth-dependent angles (Fig. 4). At the preset field-center position, median angular contrast (standard deviation/mean) rises from 0.58 to 1.53, while mean raw cosine similarity falls from 0.83 to 0.41. For nonnegative angular responses, stronger depth-selective modulation reduces the shared mean component relative to the fluctuations; the supplement gives a simple decomposition. Over 44×44 interior positions using the same 32 directions, mean absolute correlation of the full detector-integrated depth signatures falls from 0.84 to 0.48.

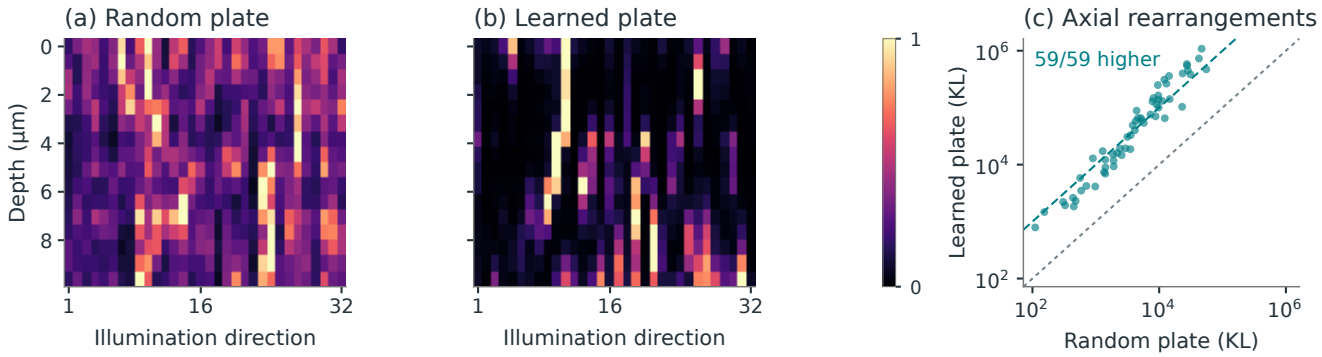


Figure 4. **Learning changes the depth code before reconstruction.** (a, b) Responses of 16 depth voxels at the preset field-center position, using one fixed camera pixel and the same 32 directions. Each row is normalized by its own maximum. The learned plate concentrates responses into depth-dependent directions. (c) Poisson measurement divergence between each neuronal volume and a merged-depth alternative preserving its projection and column centroids. Each point averages the matched direction subsets of one eligible crop; the gray diagonal indicates equal separation and the teal dashed line indicates a tenfold increase. These specified alternatives test measurement discrimination independently of a decoder.

For 59 eligible neuronal crops, we construct merged axial profiles preserving their projection and column centroids. With a global brightness scale fitted out, the learned plate increases Poisson measurement divergence from these alternatives by a factor of 10.0 (95 % CI: 8.7–11.8), increasing it on every crop. This decoder-independent diagnostic concerns specified alternatives, not uniqueness among all volumes.

6.2. Axial recovery across reconstruction priors

At eightfold measurement compression, the learned plate lowers mean axial-profile error by 41 % with TV, 50 % with pretrained PnP and 28 % with neuron-trained PnP (Tab. 1). W1 improves on all 60 crops under each method. Centroid error falls by 42 %, 54 % and 30 %, respectively, while PSNR rises by 1.6–1.8 dB. Gains persist on both prespecified and later random crops, and a second optical sampling seed gives similar results (supplement).

Figure 1 shows reduced axial smearing along a descending neurite with pretrained PnP. The supplement provides all-prior views of this crop and a reference-selected companion from another animal, together with depth slabs and a weak feature missed by both plates. Both examples have typical-to-lower W1 gains. Differences are smaller with the neuron-trained prior, which already suppresses much of the diffuse signal.

In the 12-crop control study with pretrained PnP, learned-plate centroid error is $0.230\,\mu\mathrm{m}$, below all three random plates (0.422–$0.438\,\mu\mathrm{m}$) and amplitude masks (0.396–$0.523\,\mu\mathrm{m}$). The initialization is therefore typical of the tested random plates.

In one cross-prior pairing, learned-plate TV attains lower mean W1 than random-plate neuron-trained PnP (0.218 versus $0.271\,\mu\mathrm{m}$, a paired 20 % reduction; 95 % CI: 16–23 %). Neuron-trained PnP with the learned plate gives the lowest mean W1 overall. The tested in-focus DMD remains near the depth-uniform reference under all three methods (W1 2.24–$2.38\,\mu\mathrm{m}$, reference $2.27\,\mu\mathrm{m}$ on the 12 control crops), illustrating the importance of the measurement code.

6.3. Recovery of overlapping axial structures

Lower average profile error need not imply separation of overlapping structures. We therefore test controlled crossing filaments with known depth gaps (Fig. 5). The smallest gap meeting the success criterion falls from 3.84 to $2.56\,\mu\mathrm{m}$ under TV and pretrained PnP. For neuron-trained PnP, it remains $3.84\,\mu\mathrm{m}$, despite improved W1 on neuronal volumes. At a $2.56\,\mu\mathrm{m}$ gap, the learned plate allows TV to retain two axial peaks where the random plate merges their intensities more strongly.

Neither plate reliably resolves the three-filament control, and the zero-gap control shows no false splitting. The gain in this two-filament task depends on the prior: it occurs under two of the three methods.

6.4. Comparison of optical learning objectives

We compare the squared-correlation objective with a smooth worst-case coherence criterion and an amplitude-marginalized depth Fisher-information surrogate, motivated by coherence-based sensing and Fisher-based optical design [6, 21, 25]. We implement these criteria on the same signatures and architecture, with matched initialization, sampling, optimizer and step budget (Tab. 2).

All three criteria improve on the random plate. Across both training seeds and all three decoders, our objective gives 13–20 % lower W1 than the tested coherence and Fisher surrogates, with paired intervals excluding zero. An uncentered squared-correlation variant is essentially tied with ours under neuron-trained PnP (0.207 versus $0.206\,\mu\mathrm{m}$ W1). Mean

	Axial-profile W1 ↓			Centroid error ↓			PSNR ↑	
Reconstruction	Random	Learned	Reduction [95 % CI]	Random	Learned	Reduction	Random	Learned
TV	0.371	0.218	41 % [38, 45]	0.259	0.150	42 %	36.70	38.29
Pretrained PnP	0.579	0.290	50 % [47, 53]	0.490	0.224	54 %	35.72	37.56
Neuron-trained PnP	0.271	0.195	28 % [26, 29]	0.197	0.139	30 %	38.01	39.60

Table 1. **The same learned plate improves all three reconstruction methods.** $C = 8$, matched nominal dose, 60 crops from six cells and three animals, averaging five direction subsets per crop. W1 and centroid errors are in μm; PSNR is in dB. Reductions are relative to the random initialization, with paired cell-bootstrap 95 % intervals.

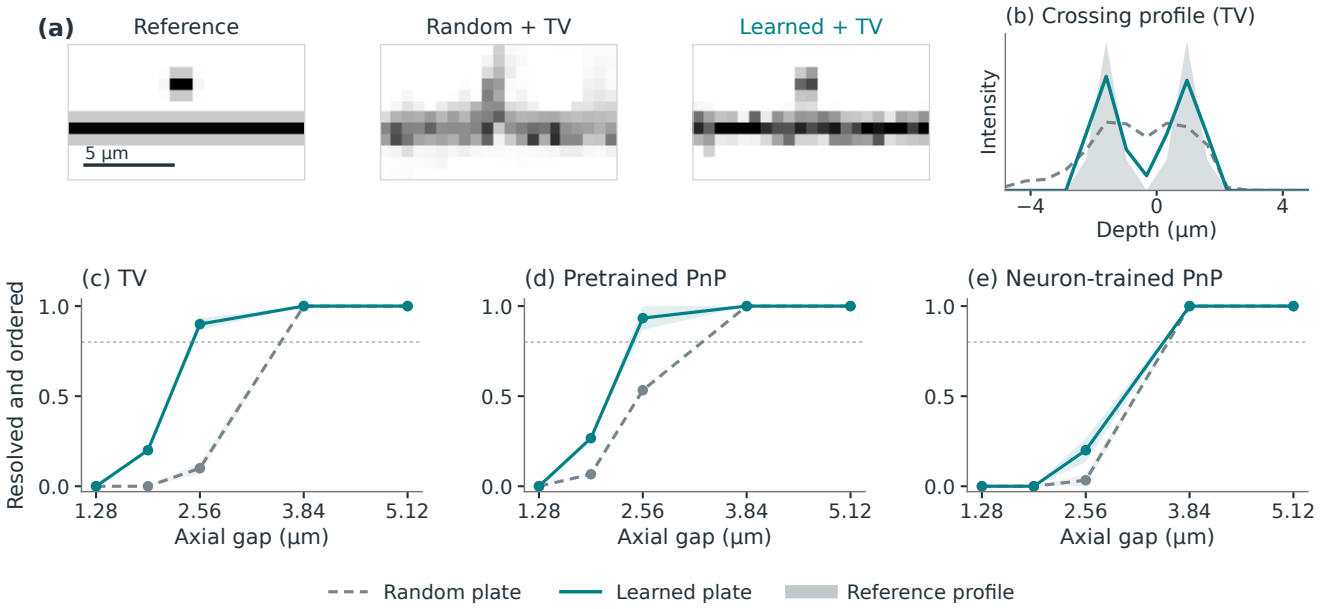


Figure 5. **Controlled axial separation is prior-dependent.** (a) Matched sections through a crossing, using one intensity scale. (b) Mean profile over its four crossing columns. The displayed gap is the learned plate's TV threshold; geometry, center and draw are the first fixed case. (c–e) Success rates across three geometries and five draws at each of two axial centers. Lines pool centers; bands span their rates, rather than confidence intervals. The 80 % threshold must hold at each center separately. Bar: 5 μm.

Optical objective	TV	Pretrained PnP	Neuron-trained PnP
Random plate	0.318	0.496	0.282
Smooth worst-case	0.238	0.346	0.238
Depth Fisher surrogate	0.228	0.340	0.240
Squared depth correlations	0.196	0.277	0.206

Table 2. **Matched optical training budgets.** Mean axial-profile W1 (μm) on the 12 prespecified crops at $C = 8$. All objectives use the same initial phase, sampled positions and angles, and 1000 Adam steps. Each encoder is tuned separately on validation data. The displayed training seed is 0; paired intervals, centroid errors and the second seed are in the supplement.

	Random plate		Learned plate
Reconstruction	256 images	32 images	32 images
TV	0.249	0.318	0.196
Pretrained PnP	0.413	0.496	0.277
Neuron-trained PnP	0.203	0.282	0.206

Table 3. **Fewer measurements with learned illumination.** Mean W1 (μm) on the same 12 crops at matched nominal dose. The learned plate uses 32 images ($C = 8$); the random plate uses either 256 ($C = 1$) or 32. Full compression and dose curves, centroids and paired intervals are in the supplement.

removal is therefore not an established source of the gain. Exact surrogate definitions and secondary results are in the supplement.

6.5. Measurement and noise budgets

The learned plate improves recovery at all three tested measurement budgets. With 32 images, it gives lower W1 than the random plate with 256 images under TV and pretrained

PnP (Tab. 3). Neuron-trained PnP gives close means (0.206 versus $0.203\,\mu\mathrm{m}$), without evidence establishing equivalence. Fewer directions use longer exposure per image at fixed nominal dose.

At 0.3 times the nominal dose, the learned plate gives 22–39 % lower W1 than the random plate at full dose across the three methods. Gains also persist when both encoders use one tenth of the dose and with added fluorescence background and read noise. Full curves and condition-specific validation settings are in the supplement.

6.6. Alignment tolerance and calibration

Accurate optical registration is important for the learned code. Under neuron-trained PnP at $C = 8$, plate-distance offsets of $\pm 0.5\,\mu\mathrm{m}$ retain 27–28 % W1 reductions when the operator is rebuilt using the known offset and scalar reconstruction parameters are revalidated. The plate itself is not retrained. Pointing errors of 0.005–0.01 in direction cosine retain 13–20 % reductions with the nominal operator.

Unmodeled registration errors can reverse the benefit. With a nominal operator, the same distance offsets make learned-plate W1 4–6 % worse than the random plate; a lateral shift of one plate feature makes it 11 % worse (95 % CI: 2–23 %). The gain therefore depends on optical registration and its representation in the operator.

7. Discussion and limitations

Learning the measurements. The same plate improves axial recovery under TV and two PnP priors, supporting optical design that leaves prior choice outside training. Prior choice still matters: neuron-trained PnP gives the lowest mean W1 with the learned plate, and all three methods use ADMM with different prior operators. These results concern the tested methods. We do not compare end-to-end learning or establish transfer to arbitrary decoders.

What improves in the recovered volume. W1 and centroid error both decrease, with similar relative gains. This supports improved axial placement without showing a larger benefit to profile shape than to its centroid. Controlled crossings establish separation gains under two priors, but no threshold gain under neuron-trained PnP. General neuronal layer separation or connectivity is not established; supplementary measurement-consistency bounds remain uninformative on dense neuronal windows.

Physical implementation. All results simulate linear fluorescence in a homogeneous medium with idealized optics and a common excitation/emission wavelength. Finer-grid measurement generation retains specimen discretization error but shares optical physics with reconstruction. Scattering, aberrations, spectral differences and unknown system calibration remain untested. The alignment results make characterization of the realized patterns and detection response a requirement for physical validation.

Extent of the evidence. Six cells from three animals limit biological generalization. Two optical sampling seeds share one initial phase; the additional random-plate and mask controls do not replicate learning from other starts. Objective comparisons use a matched step budget for specific surrogates, not an exhaustive search over design criteria. These boundaries define the scope of the optical improvement.

8. Conclusion

We learn a fixed phase plate for angle-scanned, fixed-focus fluorescence imaging from detector-integrated depth signatures. In simulations, it reduces axial-profile error by 28–50 % and centroid error by 30–54 % relative to its random initialization across TV and two PnP methods. Controlled crossings show improved separation under two priors. The results demonstrate that optical learning can improve axial recovery across the tested reconstruction methods without including them in training.

LLM Usage Statement

OpenAI ChatGPT/Codex and Anthropic Claude assisted throughout the manuscript and supplement with organization, drafting and editing, literature and code checks, numerical audits of existing simulation outputs, and figure/table code. The authors retain responsibility for the methods, results, citations and final manuscript.

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